COSAT Cosmology: The Dual-Shell Cyclic Universe a 6D Topological Framework for Macroscopic Gravity, the Emergent Cosmic Dipole, and the Big Bounce

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Abstract

Modern cosmology is constrained by the ΛCDM model, which relies heavily on unobserved phenomena to reconcile macroscopic kinematic observations with General Relativity. COSAT resolves these systemic discrepancies by re-evaluating the fundamental geometry of the vacuum. Rather than an expanding 3D kinematic void, COSAT models the universe as a highly pressurized, elastic 6D fluid metric, mathematically defined as a pseudo-Euclidean *SO(3,3)* complex phase-space manifold. By applying a fundamental thermodynamic impedance to the vacuum (the phase-damping ratio α), the metric naturally subdivides into orthogonal, phase-shifted shells. In this framework, gravity emerges not as a fundamental 3D force, but as the 10^{-36} geometric phase-leakage of the electromagnetic tensor across the Real $(+i)$ and Imaginary $(-i)$ domains. COSAT demonstrates that the observable universe is not undergoing kinematic expansion, but rather a thermodynamic free-fall toward a centralized topological cavitation void. This steepening fluid density gradient perfectly replicates the kinematic illusion of Dark Energy via tidal acceleration. Furthermore, COSAT establishes the "Big Bounce" as a localized, caustic intersection of a hyper-massive Dual-Core binary, where inside-out annihilation naturally resets the cosmic cycle. Ultimately, COSAT unites relativistic

kinematics, fluid dynamics, and quantum discrete interactions under a single, deterministic, and experimentally falsifiable topological framework.

I. Introduction: The Crisis in Standard Cosmology

Modern cosmology, governed by the ΛCDM model, is facing an existential crisis. The systemic discrepancies in the Hubble constant (the Hubble Tension), the discovery of impossibly mature galaxies by the JWST, the elusive nature of Dark Matter and Dark Energy, and the unresolved Baryon Asymmetry problem all point to a fundamental flaw in our foundational assumptions. Standard physics attempts to resolve these issues by introducing increasingly complex invisible entities and kinematic expansions.

The Complex Orbital Shell Architecture Theory (COSAT) proposes a paradigm shift: the universe is not an expanding kinematic balloon, but a highly pressurized, elastic 6D fluid metric seeking its minimum potential energy state. By modeling the vacuum as a pseudo-Euclidean *SO(3,3)* complex phase-space manifold, COSAT demonstrates that macroscopic relativistic mechanics, fluid dynamics, and quantum discrete interactions are mathematically equivalent. In this framework, the dark sector is eliminated, replaced by the deterministic structural tension of a collapsing dual-shell geometry.

II. Geometric Mechanics of the 6D Lattice

1. The Phase-Damping Ratio α

The 6D lattice is structured by thermodynamic phase-limits. The Fine Structure Constant $\alpha \approx 1/137$ acts as the fundamental thermodynamic impedance of the vacuum, dictating the subdivision of the metric into orthogonal, phase-shifted shells. The observable universe resides on the $(+i)$ outer shell, while its mirror anti-unobservable counterpart resides on the $(-i)$ inner shell. Crucially, these designations refer to the topological address of the spacetime itself, not

the charge of the particles within them; both shells contain complex distributions of matter and antimatter.

2. Gravity as Phase-Leakage

Gravity is not a fundamental force acting within 3D space. Instead, it is the microscopic geometric residue of electromagnetism leaking across the orthogonal phase-boundary into the Imaginary bulk. Because the projection angle between the electromagnetic tensor and the gravitational tensor approaches $90°$ the resulting force is roughly 10^{-36} times weaker, perfectly accounting for the Hierarchy Problem.

3. Electro-Topological Tensegrity and the Dynamic Flux Network

The stability of the dual-shell universe relies on a macroscopic tensegrity structure. While geometric Anti-Curvature actively pushes the $+i$ and $-i$ shells apart, the 10^{36} times stronger electromagnetic force actively pulls them together.

However, the 1D topological tethers connecting these domains are not rigid, 1-to-1 pairs. They form a **Dynamic Topological Flux Network**. A single proton in the $+i$ shell exists in a topological superposition of tethers—simultaneously and dynamically entangling with $+i$ electrons in its own shell, while its phase-signature projects across the boundary to pull against a fluid cloud of $-i$ antiprotons. This fluid, quantum-mechanical network of intra-shell and inter-shell flux tubes gives the 6D metric its high-pressure elasticity and provides the exact physical geometry underlying Quantum Entanglement and virtual particle interactions.

III. Completing Einstein: Anti-Curvature and the Fluid Metric

While standard General Relativity provides highly accurate predictions for localized solar system dynamics, its macroscopic application has resulted in predictive failures. COSAT asserts these failures stem from the historical mathematical simplifications used to solve the field

equations.

1. Anti-Curvature and the Cosmological Constant (Λ)

Einstein introduced Λ as an ad hoc repulsive force to prevent cosmic collapse. COSAT eliminates this arbitrary constant. In the $SO(3,3)$ manifold, macroscopic metric repulsion is generated deterministically by **Anti-Curvature**. Because the observable $(+i)$ shell and the mirror $(-i)$ shell reside on axes with opposite metric signs, their topological twist-tension projects inversely. Macroscopic repulsion is the fundamental geometric phase-pressure of the dual-shell architecture pushing against itself.

2. Reinterpreting Imaginary Solutions

In classical GR, imaginary numbers are traditionally interpreted as a breakdown of the theory. Under COSAT, imaginary solutions are valid topological coordinates. An imaginary output simply indicates that the local spatial curvature has exceeded a $90°$ phase angle relative to the observer, transitioning from the 3D Real-phase observable shell into the 6D Imaginary bulk.

IV. Grand Unification and the Topological Retardation Force

Standard physics artificially separates nature into four fundamental forces, struggling to unite the quantum scale (Strong/Weak/EM) with the macroscopic scale (Gravity). COSAT achieves Grand Unification by redefining all forces as scale-dependent expressions of the same continuous $SO(3,3)$ fluid tension.

1. The Unified Fluid Mechanics

Electromagnetism: Transverse phase-waves propagating purely along the surface topology of the $(+i)$ shell and the $(-i)$ $shell$.

Gravity: The macroscopic, geometric $90°$ phase-leakage of the electromagnetic tensor into the $(+i)$ and the $(-i)$ bulk.

The Strong Nuclear Force: The extreme, localized rigid-body tension of the 1D topological tethers. When discrete particle knots are compressed to the geometric limits of the metric, the tethers rigidly resist phase-shear, binding the nucleus.

The Weak Nuclear Force: A localized topological fracture. When local metric tension exceeds the tether's tolerance, the knot "snaps" and undergoes a discrete phase-rotation, emitting radiation and changing the particle's identity (beta decay).

2. The Fifth Force: Phase-Drag Asymmetry (Topological Retardation)

COSAT introduces a fundamental new mechanical force arising from the thermodynamic elasticity of the vacuum. Because the 6D lattice is a temporally retarded medium, geometric tension does not propagate infinitely fast. When a complex topological particle acts as a harmonic oscillator, it emits continuous energy waves. However, due to relativistic time delay, the metric's response is asymmetrical. The fluid tension in the forward vector of motion compresses, while the retarded waves in the wake arrive delayed. This generates a net, unidirectional thermodynamic force—**The Topological Retardation Force**. This ubiquitous phase-drag is the physical engine behind inertial mass and the fundamental driver of the thermodynamic arrow of time.

3. The Engine of Cosmic Evolution (Macroscopic Entropy)

This 5th force is the fundamental driver of macroscopic cosmic evolution. In COSAT, entropy is a physical, mechanical force. As the hyper-massive $(+i)$ and $(-i)$ Anti-Core orbit one another, this ubiquitous 6D phase-drag continuously bleeds their kinetic energy into the Imaginary bulk as phase-phonons. This force acts as the ultimate cosmic brake pedal, guaranteeing the deterministic orbital decay of the Cosmic Dipole.

4. Temporal Evolution of the Retardation Force (The Entropy Curve)

The strength of the Topological Retardation Force is not static; it scales dynamically with the localized density of the 6D fluid metric.

The Early Cycle (Low Drag): Immediately following the Big Bounce, the metric is highly expanded and the fluid density gradient is shallow. The resulting phase-drag is minimal.

The Late Cycle (Exponential Drag): As the $(+i)$ and the $(-i)$ shells accelerate in their thermodynamic free-fall toward the central Cavitation Void, they encounter a progressively steepening metric density. Concurrently, as the Dual-Core orbit decays, the frequency of their interaction increases. This causes the topological phase-drag to scale exponentially. Macroscopic entropy follows a sharp upward curve, resulting in a rapid, catastrophic loss of kinetic energy just prior to the Caustic Intersection.

V. Topological Annealing and the Emergent Dual-Core Oscillator

A physically robust model must naturally emerge from fundamental thermodynamic laws rather than fine-tuned initial conditions.

1. The Chaotic Plasma and Topological Annealing

When the cosmic cycle resets, the local metric is shattered into a chaotic plasma of $(+i)$ and $(-i)$ micro-cores. As this plasma cools, it seeks its lowest potential energy state through Topological Annealing. Governed by coalescence (gravity), anti-curvature repulsion, and topological tethering, the system naturally freezes into a **Cosmic Dipole**. The $(+i)$ mass coalesces into a singular hyper-massive Core, while the $(-i)$ mass forms a mirror Anti-Core.

2. The Dual-Core Binary Engine

These two mutually repelling nodes orbit the central barycenter. As they spin, they act as a colossal topological propeller, generating alternating transverse phase-waves that dictate the macroscopic "beat pattern" of the Cosmic Web, sweeping galaxies into sprawling filaments.

VI. The Caustic Intersection and the Gamma-Ray Detonation

In standard cosmology, the uniform temperature of the observable universe necessitated the invention of Cosmic Inflation. COSAT eliminates the need for inflation by re-establishing the "Big Bang" as a literal, centralized thermodynamic detonation occurring within a pre-existing, causally connected fluid metric.

1. The Topological Caustic and Inside-Out Annihilation

As the Dual-Core orbit decays, the cores spiral inward until they intersect at the zero-gravity L1 barycenter. This forms a **Caustic** —a localized geometric boundary where structural tension diverges to infinity. As the cores crush together, extreme topological phase-shear forces the Real and Anti-Real mass components into the same geometric space.

2. Radiation Pressure and the Cavitation Void

This triggers a catastrophic matter-antimatter annihilation at the core's center, converting the mass entirely into ultra-high-energy photons. The resulting extreme **Radiation Pressure** vastly exceeds the ambient containment tension of the 6D lattice. This supreme gamma-ray detonation blasts the surviving outer plasma radially outward, leaving behind a localized, near-perfect metric vacuum—the Topological Cavitation Void.

3. The Resolution of the Inflation Paradox

Because the material of the universe was physically compressed into this singular caustic binary *prior* to the detonation, it was already in causal thermal equilibrium. When radiation pressure blew the cores apart, the expanding plasma was naturally uniform in temperature, requiring no hyper-luminal inflation.

VII. The Kinematic Illusion of Dark Energy

1. The Thermodynamic Free-Fall

The universe is not expanding. Following the outward blast, the ambient pressure of the outer 6D lattice halts the plasma's expansion. Both the $(+i)$ and the $(-i)$ shells then begin to free-fall back toward the central Cavitation Void, moving down a steepening fluid density gradient.

2. The Illusion of Cosmological Redshift

Light traveling from distant galaxies to an observer in either the $(+i)$ or the $(-i)$ shells is climbing out of a deeper gravitational well, resulting in extreme, distance-dependent macroscopic gravitational redshift. Because the gradient steepens toward the core, galaxies further ahead in the free-fall accelerate away from observers due to tidal forces—creating the exact kinematic illusion currently misidentified as Dark Energy.

3. The Singularity Core Radius (SCR)

The observable boundary of the universe (the Hubble Radius) is not a kinematic speed limit. It is the Singularity Core Radius (SCR)—the macroscopic event horizon where gravitational redshift becomes infinite and matter phase-rotates $90°$ into the Imaginary bulk (Evanescence).

VIII. Boundary Thermodynamics and Phase-Shear Radiation

As matter is crushed through the $90°$ Evanescence boundary at the SCR (or local black hole horizons), it experiences unimaginable Topological Phase-Shear. The complex knots of standard baryonic matter are ripped apart into pure, luminal $(+i)$ and $(-i)$ particle pairs. This violent geometric friction generates intense phase-phonons, leaking high-energy Planck-scale radiation and Ultra-High-Energy Cosmic Rays (UHECRs) back into the observable shell as thermodynamic exhaust.

IX. Proposed Experimental Verifications

COSAT provides specific, falsifiable predictions that diverge from standard General Relativity:

1. Intra-Shell Antimatter Gravitational Symmetry

Standard physics struggles to definitively prove why antimatter and matter behave identically under gravity. COSAT explicitly predicts this via the Complex Phase Plane. Because gravitational projection is dictated strictly by the geometric shell (i-axis) and not electromagnetic charge (R-axis), local $(-R, +i)$ antimatter synthesized in a laboratory will experience the exact same geometric phase-leakage as standard $(+R, +i)$ matter. Recent drop tests (such as CERN's ALPHA-g) confirming standard downward acceleration perfectly validate that the R-axis does not influence metric curvature.

2. Macroscopic Anti-Curvature Lensing (The $-i$ Signature) Because observers in the $+i$ shell cannot physically obtain a sample of $-i$ matter, the mirror shell must be verified macroscopically. Clumps of $-i$ mass in the mirror shell will project inverse geometric phase-leakage (Anti-Curvature) into our observable $+i$ shell. Astronomers should look for anomalous "divergent" gravitational lensing—regions of deep space that act as gravitational magnifying glasses, but bend light *outward* rather than inward, indicating the presence of a massive $-i$ structure pushing against our local metric.

3. Gravitational Particle Oscillation Particles that naturally oscillate between matter and antimatter states (e.g., Kaons, B-mesons) should experience a shift in their oscillation frequency when placed in a strong gravitational field, as the "clock" of the alternate state ticks at a slightly different rate due to the phase-shear of the gravity well.

X. Conclusion

COSAT unites relativistic kinematics, fluid dynamics, and quantum mechanics under a

single deterministic framework. The universe requires no dark sector, no miraculous expansion, and no thermal death. It is a continuous, perpetual topological hourglass—a self-organizing Dual-Core oscillator driven by the conservation of geometric tension within a 6D complex phase-space.

XI. Applied Mechanics & Topological Engineering (Appendices)

Appendix A: Mathematical Formalism & The COSAT Transformation Tensor

To enforce the geometric tension across both domains, the vacuum operates under the *S*O(3,3) metric tensor η. We define the Complex Phase-Interval Invariant $(\mathcal{J})$ as:

$$d\mathcal{J}^2 = d\boldsymbol{R}^2 - d\mathbf{I}^2$$

The COSAT Transformation Tensor ($\mathbb{T}$) dictates all physical interactions, crossing spatial translation Λ with complex phase-rotation (Θ) :

$$\mathbb{T} = \Lambda \otimes \Theta$$

$$\Theta(\theta) = \begin{bmatrix} cosh(\theta) \cdot \mathrm{I}_3 & sinh(\theta) \cdot \mathrm{I}_3 \\ sinh(\theta) \cdot \mathrm{I}_3 & cosh(\theta) \cdot \mathrm{I}_3 \end{bmatrix}$$

Applying this tensor yields the foundational equations of COSAT:

$$\boldsymbol{R}' = \boldsymbol{R} cosh(\theta) + \mathbf{I} sinh(\theta)$$

$$\boldsymbol{I}' = \boldsymbol{R} sinh(\theta) + \mathbf{I}\, cosh(\theta)$$

By equating geometric phase-angle (θ) to standard rapidity, where $cosh(\theta) = \gamma$, the equation perfectly mirrors the Lorentz Transformation.

The COSAT Complex Phase Plane

Under COSAT, every coordinate in the metric is defined by a complex phase state (R, i), where R dictates Electromagnetic Charge (the minor axis) and i dictates the Geometric Shell (the

major axis). This maps four topological quadrants: Real Matter $(+R, +i)$, Real Antimatter $(-R, +i)$, Mirror Matter $(+R, -i)$, and Mirror Antimatter $(-R, -i)$.

- **Minor Annihilation:** Crossing the R axis within the same shell (standard particle-antiparticle annihilation), releasing radiation without breaking the local metric.
- **Major Annihilation:** Crossing the i axis (Caustic Intersection), where the geometric shells themselves perfectly cancel $(+i + (-i) = 0)$, triggering a universal thermodynamic reset.

The Topological Phase Cone (Visual Geometry)

The COSAT Phase Plane can be visually mapped as a geometric cross-section resembling a four-leaf clover bisected by an "X" (a modified Minkowski diagram):

- **The "X" (Null Geodesics):** These diagonal lines represent the luminal boundary ($v = c$). They act as impenetrable asymptotic walls of metric tension.
- **The Four Leaves:** These discrete regions represent the four topological quadrants.
- **The Origin** $(0,0)$**:** The exact center where the axes intersect, representing a state of zero rest mass and zero charge (the photon).

Because mass cannot accelerate past the speed of light to cross the "X" boundary, standard matter is strictly confined to its native topological leaf. The only geometric mechanism capable of crossing from the $+i$ shell to the $-i$ shell is a Major Annihilation at the exact $(0,0)$ origin. Here, mass is converted into luminal radiation, allowing the energy to safely ride the boundary geodesics into the opposing metric.

The $2\times$ Luminal Deflection Proof

When a photon passes a massive body (e.g., the Sun), standard General Relativity dictates a $2\times$ deflection compared to classical Newtonian gravity. COSAT derives this $2\times$ limit natively via topological superposition: At the luminal boundary ($v = c$), 3D space collapses into a 2D plane. The $(0,0)$ photon interacts with the superposition of all four geometric "leaves" of the phase plane. While this creates four topological degrees of freedom ($4\times$ the baseline geometric tension), half of this tension is parallel to the photon's momentum vector. The photon only experiences the orthogonal, transverse phase-drag from two of the four quadrants. This $2\times$ transverse geometric shear exactly replicates the established macroscopic lensing effect without violating the conservation of energy.

Appendix B: Calibration of the 6D Metric Density via the SCR Horizon

Treating the Hubble Radius $1.36\times10^{-26}m$ as the gravitational Evanescence boundary (SCR), we reverse-engineer the required structural density of the 6D lattice. Setting the denominator of the Schwarzschild macroscopic redshift formula to zero yields the critical radius r_{crit}:

$$1-\sqrt{\frac{8\pi G\rho r^2}{3c^2}}=0 \quad\Longrightarrow\quad r_{crit}=\sqrt{\frac{3c^2}{8\pi G\rho}}$$

Solving for ρ:

$$\rho=\frac{3c^2}{8\pi G(r_{crit})}\approx 8.6\times10^{-27}kg/m^3$$

This perfectly matches the standard ΛCDM Critical Density requirement. The "missing 95%" Dark Matter is entirely accounted for by the baseline structural phase-pressure of the $SO(3+3)$ lattice.

COSAT Cosmology 2.0: From The Hydrodynamic Eternal Engine To the possibility of a Quantized 6D Harmonic Multiplex

Proposed by Donald Cameron (March 2026)

Abstract

COSAT 2.0 proposes a shift from static cosmological models to a **deterministic, self-regulating 6D harmonic engine**. By replacing kinematic expansion with a **topological heartbeat**, this model addresses the "Crisis in Cosmology" and baryon asymmetry through the mechanics of a **Recurrent Caustic Intersection**. In this model the universe is represented by a **quantized 6D spherical standing wave function** within a pressurized 6D manifold, where traditional "Big Bang" events are redefined as **Multi-Frequency Antineutrino Laser Bounces**. These periodic detonations re-establish metric tension, triggering a **Diastolic Free-Fall**—the phenomenon currently interpreted as expansion. The paper explores the possibility of a **Multiplex Symphony**, where discrete "bubble universes" reside as harmonic oscillators between nodal zero-geodesics. These boundaries act as **Annihilation and Evanescence sources**, facilitating phase-tunneling across potential energy barriers. By interpreting gravity and inertia as emergent properties of **topological phase-drag**, COSAT 2.0 provides a "Bass Awkward" perspective from which the Cosmic Microwave Background is thought to be the **terminal output** of a previous cycle. This framework invites further inquiry into the quantization of field lines as real positive and negative standing waves, offering a mechanical alternative to Dark Matter, Dark Energy, and Inflationary theory.

Note: The COSAT framework is designed as a living, iterative topological model. The following addendum represents an evolutionary refinement to the immediate post-bounce timeline established in Section VI, demonstrating the framework's capacity to elegantly resolve early-universe observations through macroscopic fluid mechanics rather than forced initial conditions.

The Evolutionary Sequence (The Hydrodynamic Engine)

1. **The Caustic Intersection & SG Siphon:** Dual-Cores collide at the (0,0) origin, triggering an inside-out annihilation. A macroscopic **Stern-Gerlach field** siphons boundary-synthesized antimatter into the core, triggering a secondary exponential cascade which separates the un-tethered $+R$ and $-R$ matter into distinct phase-regions. Only after this massive thermodynamic sorting does the universe cool sufficiently for the topological tethers to lock, forming the stable, matter-dominated $(+i)$ nodes of our observable region and the stable $(-i)$ nodes of the internal region that is repelling our region. Note that relative to observers in the $(-i)$ region our region appears to be an internal region.
2. **Acoustic Shrapnel & The Antineutrino Laser:** The Big Bounce shatters the metric into a power-law distribution of repelling (+i) and $(-i)$ **micro-cores**. A coherent pulse of antineutrinos in the $(+i)$ region we call the observable universe and a corresponding cohehrent pulse of neutrinos in the $(-i)$ region that is inner relative to our own—

an **antineutrino laser in the** $(+i)$ region and a neutrino laser in the $(-i$ vibrates and dissolves unstable micro-fragments in each region, re-injecting dense fluid into the expanding metric.

3. **The Thermodynamic White Hole**

   Because the Big Bounce implosion leads to the formation of a **quark-antiquark plasma** and ultimately to a localized untethered metric inward and outward detonation state, its macroscopic evolution appears to be mathematically identical to a time-reversed black hole (a White Hole) relative to observes in their respective imaginary regions. As the fluid metric expands and cools, it naturally cascades through specific, high-energy state transitions.

Transitions: As the metric cools, quarks bind, forming a hyper-dense outer and inner repelling regions containing degenerate states of separated neutron and anti-neutron matter which decay in response to the fluid surge from the evaporating micro cores, the metric transitions into a region containing primarily a **Proton/Electron Plasma** consisting of highly excited protons and unbound electrons and an region containing an **Anti-Proton/Anti-Electron Plasma** consisting of highly excited Anti-protons and unbound Anti electrons, that upon further cooling produced a set of nuclear, atomic and molecular population inversions i.e., the formation of Helium nuclei, heavy isotopes of hydrogen and perhaps several other of the lighter elements and finally molecular hydrogen; which all combined to produce a coherent multifrequency flash consisting primarily of Gamma, UV, Visible, and IR frequencies of electromagnetic radiation, which is the origin of the CMB and are responsible for the **Holographic Speckling** anisotropies in the CMB which represent the **interference pattern** of the multi-frequency laser flash interacting with the hydrodynamic wakes of the micro-cores.

Standard cosmology attributes the Cosmic Microwave Background (CMB) to the random thermal scattering of photons during Recombination. COSAT proposes a radically different, highly synchronized mechanical origin. Because the universe is expanding as a unified geometric shell, the specific temperature and pressure threshold required for electrons and antielectrons to couple to protons and antiprotons is reached almost simultaneously across the macroscopic $(+i)$ metric.

When these electrons and antielectrons cascade down into their ground states across the entire shell, they undergo a massive, universe-spanning **Population Inversion**. Rather than a chaotic thermal fog, this state-transition triggers a synchronized burst of stimulated emission. The early COSAT universe did not simply glow; it flashed into existence as a highly coherent **Cosmic LASER**. Over billions of years of thermodynamic free-fall, this coherent, high-energy emission has been systematically stretched by macroscopic geometric redshift into the ambient microwave static observed today. Note COSAT does not assume that this geometric redshift is due to the inflation of the universe.

4. Mathematical Unification Highlights

- **The Unified Knot Force:** This encapsulates the Strong Force (tether tension), Electroweak Force (geometric twisting/fracturing), and Local Gravity (residual phase-leakage). Gluons are eliminated; nuclear binding is defined as the physical braiding of 1D tethers.

- **Dimensional Transformation:** Transformations between 1D (tethers) and 6D (bulk) frames utilize **hyperbolic trigonometry** (*tanh* for velocity and *cosh* for mass/time dilation) to manage the infinite density of the 45° luminal boundary.
- **The PMV Vector:** Time and antimatter are redefined as kinematic differential states along the **Primary Momentum Vector** (entropy axis) rather than separate dimensions.
- **The Asymmetry Force (Thermodynamic Drift):** A macroscopic force driven by the **Relativistic Delay** of waves in an asymmetrically expanding potential well. This siphons mass toward the region of a particle's boundary that appears to be older, colder, more red-shifted, creating the kinematic illusion that there exists in the observable universe a mysterious attractive force that is inversely proportional to the size of the observable universe.
- **No need for Dark matter:** The flattening of galactic rotational curves, and gravitational lensing are explained by the introduction of a density times volume parameter such that all of the atoms in the space enclosed by a sphere with radius equal to the distance between two objects are included in determining the gravitational force using either Newton's law or the Schwarzschild solution to the Einsteinian field equations. Thus, no dark matter or ad hoc modification to gravitational theory are needed. Additionally the magnitude of what used to be called the gravitational force needs to be doubled because in Newtonian physics, gravity is only proportional to the mass of the interacting particles affects mass but in 1919, Arthur Eddington proved during a solar eclipse that light bends exactly **twice as much** around a star as Newtonian math predicts. Mainstream GR explains this by saying light is affected by both the curvature of space *and* the curvature of time. **In COSAT theory** this doubling happens because photons "travel" on the luminal boundary which is the absolute edge between the real sub-luminal axis and the imaginary phase space which means it experiences the topological pressure of the dense fluid (gravity) at double the standard rate of a slow-moving, heavily knotted object. It is paying a massive thermodynamic tax just to exist in that boundary state. In Relativity, light travels along a **null geodesic** along which time dilation is infinite and length contraction in the direction of its momentum is infinite. Thus, from the photon's frame of reference the distance between a source and a receiver is zero. In COSAT theory the photon does not simply "travel" between a source and a receiver. The source knot (a star) and the receiver knot (our telescope) are in superposition. They are physically touching via a shared 1D tether along the Imaginary axis. The photon is simply the instantaneous transfer of kinetic energy from one knot to another along that tether.

  Great Attractors, Cosmic Voids and Galactic formation

  This model offers several possible mechanisms for the creation of cosmic voids and great attractors as well as the creation of galaxies, including galaxies that the JWTS recently revealed to be in places they should not exists on the cosmic evolutionary timeline. To make a long story short these mechanisms have to do with the size and nature of the shards from the original annihilation driven uncoupling event. Depending upon which scenario one choses to use to create the great attractors and the cosmic voids one will be able to use gravitational lensing and asymmetries in the Hubble flow to distinguish between them. For example, in one model in which the cosmic voids were assumed to be created by tunneling events early in the evolution of the universe in which negative imaginary black holes briefly tunneled into the positive imaginary region of space we call the observable universe and before being pushed back out repelled large amounts of

positive imaginary matter which created cosmic voids that do not significantly lens light. Contrast this with the results of another model in which in which it was assumed that very large slow moving black hole shards pulled in large amounts of positive matter which created voids; a scenario in which these types of voids would lens light passing around them. Additionally, a third model was created in which antimatter of the positive imaginary type tunneled across the boundary early in the history and was annihilated leaving a large void. In this model as in the first model such voids would not exhibit a significant amount of gravitational lensing. The question that needs to be addressed is are all three of these types of voids possible in the observable universe?

Epilogue: An Open Invitation to the Scientific Community

The Framework as a Sandbox, Not a Dogma

The Complex Orbital Shell Architecture Theory (COSAT) is presented here not as an absolute, unassailable truth, but as a foundational architectural blueprint. History has shown that when theoretical models are treated as rigid dogma—new "six-day creation cycles"—they stall the progress of human understanding. COSAT is designed to be the exact opposite: an open-source thermodynamic sandbox.

It is my hope that this framework serves as a launchpad for the creativity, rigorous testing, and mathematical refinement of others. There are profound, macroscopic questions left to be explored within the 6D fluid metric. For example:

- **The Global Phase-Rotation (The Quadrillion-Year Inversion):** Does the entire $SO(3,3)$ manifold possess a macroscopic angular momentum? If the structural hourglass is slowly rotating through the complex plane, a full $180°$ geometric phase-shift over quadrillions of years would mathematically invert the metric ($i^2 = -1$). The observable shell would seamlessly rotate into the Imaginary bulk; up would become down, and inside would become outside, perpetually flipping the cosmic topology.
- **The Granularity of the Retardation Force:** How does the specific fluid density of the metric at the quantum scale dictate the exact half-lives of standard model particles during the Thermodynamic Free-Fall?
- **The Un-Conjugated Electro-Gravitational Cross-Term:** In standard quantum mechanics, the complex scalar of a state is derived via its complex conjugate (ZZ^*), which artificially erases the cross-terms. However, if the $SO(3,3)$ metric does not inherently conjugate the thermodynamic interaction between mass and charge, the literal expansion of the electro-gravitational state yields $(q + im)^2 = q^2 - m^2 + 2iqm$.

  (A Note on Dimensionality: In standard SI units, adding charge (q) to mass (m) is dimensionally impossible, as they are treated as fundamentally different properties. However, advanced theoretical physics frequently utilizes Natural Units (c = G = 1) to bridge this gap. In Phase Kinetics, this summation is natively legal without hidden constants. Because both charge and mass are defined strictly as manifestations of the exact same SO(3,3) fluid momentum flux (polarized pressure vs. unpolarized leakage), they share the exact same foundational geometric dimensions.) While q^2 represents the Real electromagnetic force and $-m^2$ represents the inverted Real gravitational equivalent, the $2iqm$ term represents an Imaginary, transverse shear force strictly requiring both charge and mass. It needs to be determined if this imaginary $2iqm$ cross-term is the literal mathematical expression of the **Topological Retardation Force** (inertial phase-drag) that results from the time delay of complex

the topological retardation force (the inertial phase drag) and the electromagnetic force in all its subtle guises are the only two forces needed to account for the evolution of the observable universe, with the evolution of the Topological retardation force being immeasurably small in the present day observable universe but very powerful when it is in its high density phase of evolution i.e., at its alpha and its omega. .

The universe is vast, and our capacity to understand it requires collaborative, iterative thought. I invite physicists, mathematicians, and topological engineers to take this framework apart, identify its limits, and build something even better.

https://github.com/irxyzcameron6D/Cosmology-1.0-

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